UniEarth: A Unified Foundation Model for Remote Sensing Image Generation and Understanding

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***Abstract*—Remote sensing (RS) underpins geospatial intelligence by enabling large-scale Earth observation (EO). Existing RS approaches are typically developed along two separate lines: multimodal understanding (MMU) and image generation. Such separation limits parameter sharing and the potential synergy between understanding and generation. A unified foundation model can in principle support both tasks, but it still yields inferior performance in the RS domain due to omitting specific issues, including RS data scarcity, granularity mismatch between RS imagery and text, and optimization instability under long-tailed distributions. To address the above issues together, we propose a unified RS foundation model, called `UniEarth`, for joint RS-oriented understanding and generation. Built on the linear-complexity Mamba-2 architecture, `UniEarth` is data-efficient even with limited RS supervision. To stabilize joint optimization under coarse RS annotations, we design a cross-modal semantic anchoring mechanism on the intermediate hidden states of the shared backbone, balancing the generation and understanding pathways. Furthermore, `UniEarth` designs difficulty-aware bandit optimization (DA-UCB) to stabilize training by adaptively prioritizing low-resource categories in long-tailed distributions. Experiments on widely used RS benchmarks show that `UniEarth` consistently outperforms state-of-the-art methods on both understanding and generation, with a 44.6% relative reduction in Fréchet inception distance (FID) for image generation and a 20.1% relative improvement in CIDEr for captioning, while also achieving leading accuracy on visual question answering and scene classification.**

***Index Terms*—Multimodal understanding (MMU), remote sensing (RS), text-to-image generation, unified foundation model.**

I. Introduction

REMOTE sensing (RS) is fundamental to geospatial intelligence, providing large-scale and continuous Earth observation (EO) for applications such as environmental monitoring [1], precision agriculture [2], urban planning [3], and disaster response [4]. UAV platforms further enable flexible EO for tasks such as object detection, tracking, and semantic segmentation [5].

Existing research in RS can be broadly divided into two lines of work. The first line focuses on multimodal understanding (MMU), utilizing large-scale image–text pairs to pretrain encoders for tasks like retrieval and captioning [6], [7], [8]. Representative models such as RemoteCLIP [9], RSGPT [10], and GeoChat [8] have demonstrated strong capabilities in cross-modal retrieval, scene classification, visual question answering, and image captioning. The second line concentrates on image generation, primarily employing diffusion-based architectures for text-to-image synthesis [11], [12]. Recent methods such as Text2Earth [12] and CRS-Diff [13] have achieved progressively higher generation fidelity by leveraging layout-guided conditioning and controllable synthesis mechanisms. Although both lines have achieved satisfactory model performance, their separation hinders cross-task knowledge transfer and limits the potential synergy between understanding and generation. For instance, discriminative features learned during understanding training could in principle guide the generation branch to produce more semantically faithful images, while generative data augmentation could in turn diversify the training distribution for understanding tasks.

A unified RS foundation model is desirable because understanding and generation are closely related in RS interpretation. Discriminative representations can provide semantic constraints for image synthesis, while generative modeling can expose the shared backbone to diverse visual patterns. However, existing RS methods usually train these two capabilities in separate pipelines, which limits cross-task interaction and increases the cost of deployment in practical RS systems. Fig. 1 provides a comparative overview of representative models across understanding and generation benchmarks.

General-domain unified multimodal models (UMMs) have recently achieved remarkable progress by integrating MMU and image generation within a single architecture [14], [15], [16], [17], [18]. A natural strategy is therefore to transfer this unified paradigm to RS. However, as illustrated in Fig. 2, such transfer is nontrivial and often suboptimal because RS data present three domain-specific challenges.

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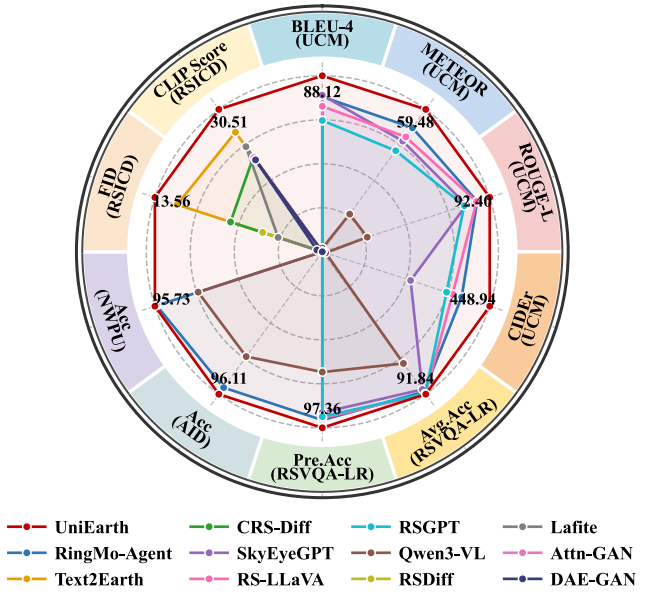


Fig. 1. Radar visualization of model performance across understanding and generation benchmarks.

First, compared with the general domain, RS training data is limited and expensive to annotate. Unified vision-language models are commonly trained on hundreds of millions to billions of image–text pairs, while the available RS image–text corpora are orders of magnitude smaller. Moreover, many RS captions provide scene-level descriptions rather than fine-grained spatial annotations, making high-quality supervision for unified training scarce.

Second, RS suffers from a supervision granularity gap between modalities. RS imagery contains fine-grained structures and many small objects, including vehicles in parking lots, ships in harbors, and buildings in dense urban areas. However, available textual supervision is often sparse and scene-level. A caption such as an urban area with residential buildings usually describes the overall scene but omits spatial layout and fine-grained object composition. This mismatch leads to under-constrained cross-modal grounding: during understanding, the model may learn coarse scene-level associations rather than object-level semantics; during generation, it may produce fine details that are weakly constrained by the text prompt.

Third, RS datasets are often long-tailed. Common categories such as vegetation, buildings, and roads dominate both image content and the training corpus, while less frequent but important categories such as harbors, stadiums, and bridges are underrepresented. Consequently, traditional joint training strategies can overfit dominant patterns and under-explore low-resource concepts, resulting in weaker generalization on underrepresented categories. This issue is exacerbated in unified training, where the generation and understanding branches may compete for optimization capacity, and the branch with more abundant supervision, usually understanding, may dominate gradient updates.

A direct way to build unified RS models is to fine-tune general-domain unified architectures on RS data. However, our experiments show that this strategy still leaves a clear performance gap, especially for generation. This indicates that RS data should be complemented with mechanisms that address coarse supervision and long-tailed distributions.

To address the above domain-specific issues together, we propose `UniEarth` via contrastive matching alignment (CMA) and difficulty-aware bandit optimization (DA-UCB) to bridge the supervision granularity gap and to better handle long-tailed data. Specifically, CMA imposes cross-modal semantic anchoring on the intermediate hidden states of the shared backbone, encouraging the generation-side representations to remain semantically consistent with the input text even under coarse RS annotations. DA-UCB formulates the training process as a multiarmed bandit problem, where difficulty clusters of training samples are treated as arms and an online upper confidence bound (UCB)-based policy dynamically allocates learning focus to balance exploration of underrepresented patterns with exploitation of well-learned knowledge. With these two components, `UniEarth` improves both understanding and generation performance using a compact RS corpus of approximately one million samples.

The main contributions are summarized as follows.

1) We propose `UniEarth`, a unified RS foundation model that jointly supports MMU and text-to-image (T2I) generation within a single next-token prediction framework. Built on the linear-complexity Mamba-2 architecture, `UniEarth` enables unified RS understanding and generation simultaneously with a compact domain-specific corpus of about one million samples. On the evaluated benchmarks, `UniEarth` consistently achieves state-of-the-art performance, reducing Fréchet Inception Distance (FID) by 44.6% and improving CIDEr by 20.1% over the previous best results.
2) To bridge the supervision granularity gap, we design CMA as a cross-modal semantic anchoring mechanism on the intermediate hidden states of the shared backbone, thereby stabilizing the generation branch under coarse RS annotations and benefiting joint optimization across both tasks.
3) To better cope with long-tailed distributions, we propose the DA-UCB strategy for `UniEarth`. By formulating training as a multiarmed bandit problem, DA-UCB provides an online policy that adaptively allocates learning focus, balancing exploration of underrepresented patterns with exploitation of well-learned knowledge.

The remainder of this article is organized as follows. Section II reviews related work on RS MMU, RS image generation, and unified vision-language models. Section III introduces the preliminaries ON state space models and multiarmed bandits. Section IV presents the proposed method, including CMA, DA-UCB, and the overall training objective. Section V describes the dataset construction and decontamination procedure. Section VI reports extensive experimental results and ablation studies. Section VII discusses limitations and future directions. Finally, Section VIII concludes this article.

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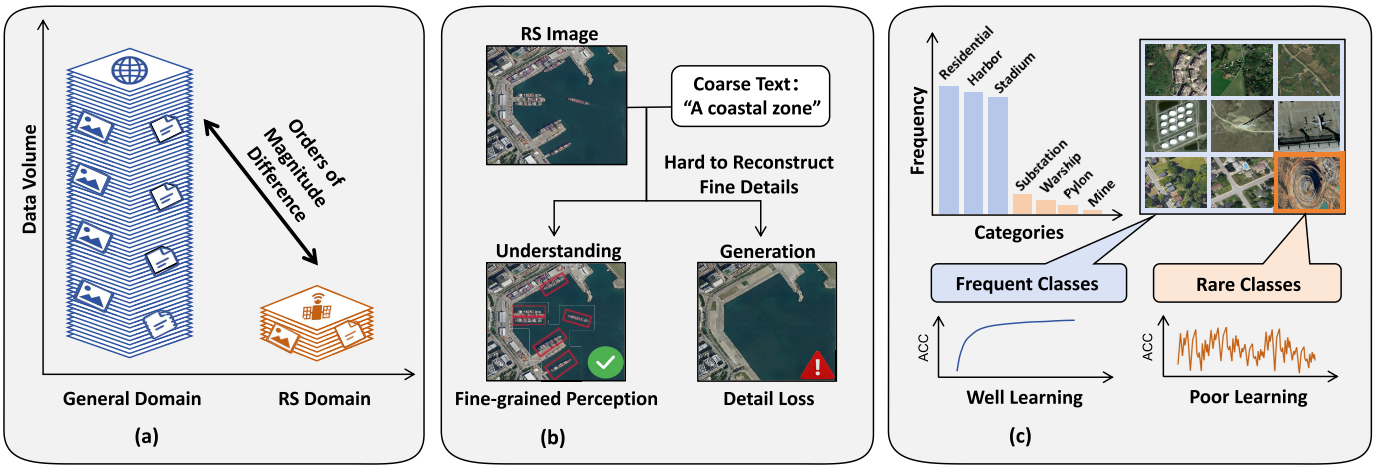


Fig. 2. Illustration of the three domain-specific issues hindering unified RS modeling. (a) Data scarcity difference between general and RS domains. (b) Granularity mismatch leads to under-constrained cross-modal grounding. (c) Optimization instability under long-tailed distributions.

II. Related Work

A. RS Multimodal Understanding

Before the emergence of multimodal foundation models, RS understanding was mainly addressed through task-specific methods. Representative studies include tracking methods for UAV videos [19], [20], [21], [22], multitarget tracking for satellite videos and video SAR [23], [24], and hierarchical classification for fine-grained high-resolution range profile target recognition [25]. Although effective in their respective scenarios, these methods are generally tailored to specific platforms and tasks, motivating the development of more general multimodal foundation models. With the rise of foundation models, RS vision-language understanding has progressed rapidly. Early efforts emphasized robust visual representation learning and cross-modal alignment. Methods such as RemoteCLIP [9] employ masked image modeling and contrastive learning to build transferable feature spaces for retrieval and zero-shot classification. SkyScript [7] further scales up vision-language pretraining by constructing a large and semantically diverse RS dataset.

Building on these representations, research has increasingly moved toward large multimodal models (LMMs) that can follow complex instructions. RSGPT [10] introduces an RS-specific vision-language model with a human-annotated benchmark for captioning and VQA. RS-LLaVA [26] adapts the LLaVA architecture to RS through low-rank adaptation, enabling joint captioning and question answering. H^2RSVLM [27] proposes a versatile vision-language model that improves factual accuracy in RS image analysis. Models including EarthGPT [6], GeoChat [8], and SkyEyeGPT [28] integrate multiplatform data and agent-style reasoning to answer open-ended queries, while RingMoGPT [29] and RingMo-Agent [30] extend the scope to include object detection, grounded captioning, and multitemporal change analysis within unified instruction-tuning frameworks. Other works study fine-grained grounding [31] to localize objects within large geospatial scenes, and RS-specific VQA methods [32], [33], [34], [35] explore bi-modal fusion and spatial hierarchical reasoning to improve question answering on RS imagery. However, these models are primarily understanding-oriented and do not natively support text-to-image generation. As a result, the semantic representations learned from perception, reasoning, and grounding tasks are not directly coupled with visual synthesis, motivating the exploration of unified architectures.

B. RS Image Generation

Image generation in RS has evolved from early GAN-based methods to modern diffusion-based architectures. GAN-based approaches, including StackGAN [36], DAE-GAN [37], and DF-GAN [38], adapt text-to-image synthesis techniques from the natural image domain to RS but often suffer from mode collapse and limited fidelity on complex overhead scenes. Txt2Img-MHN [39] introduces modern Hopfield networks to improve text–image alignment, while language-free training strategies [40] explore alternative supervision paradigms to reduce reliance on paired text data.

The advent of diffusion models has brought substantial quality improvements. RSDiff [11] adapts diffusion models for RS text-to-image generation, while Text2Earth constructs a large-scale global dataset and introduces layout-guided generation. DiffusionSat [41] develops a generative foundation model for satellite imagery with controllable synthesis, and CRS-Diff [13] adapts ControlNet-style architectures to RS for fine-grained spatial manipulation. For SAR data augmentation, Ship-Go [42] formulates ship image synthesis as an instance-to-image diffusion task and jointly controls ship instances and environmental prompts to improve both object and background diversity. CityDreamer [43] further explores compositional 3-D world modeling for urban scenes.

Beyond image synthesis, EO models such as TerraMind [44] and RemoteBAGEL [45] explore modality completion and world modeling for downstream geospatial tasks [46], whereas EarthVL [47] unifies vision-language understanding with textual generation. However, these generation-focused methods operate in isolation from understanding tasks and require large-scale paired data that is scarce in the RS domain.

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C. Unified MMU and Generation

UMMs that jointly support understanding and generation within a single architecture have become an active research direction in general-domain vision-language learning. Existing UMMs can be broadly grouped into three architectural paradigms.

The first is autoregressive (AR) architectures, which treat both text and visual tokens as a single sequence and model them through next-token prediction. Emu [48] and the SEED series [49] show that unified AR modeling over discrete visual tokens can yield generalist multimodal capabilities. Chameleon [14] scales this idea with mixed-modal early fusion training, while Janus-Pro [15] improves performance through better data curation and model scaling.

The second is diffusion-augmented architectures, which delegate image synthesis to a diffusion head while retaining AR modeling for text. Show-o [16] integrates discrete diffusion into a transformer-based unified model, and Transfusion [17] jointly predicts text tokens and diffuses image tokens within a single multimodal backbone.

The third is hybrid architectures that combine AR understanding with flow- or diffusion-based generation through separate but coupled pathways. JanusFlow [50] harmonizes AR and rectified flow objectives, and NExT-GPT [51] extends unified modeling to arbitrary-to-arbitrary generation across modalities. BLIP-2 [52] takes a different route by bridging frozen image encoders and LLMs via a lightweight querying module, establishing an influential framework for vision-language alignment.

Orthogonal to the choice of generation paradigm, the backbone architecture also plays a role in unified modeling efficiency. OmniMamba [18] replaces the Transformer backbone with state space models, reducing sequence modeling complexity from quadratic to linear. This property is attractive for RS imagery, where high spatial resolution leads to long token sequences. `UniEarth` adopts OmniMamba as its architectural starting point.

While these general-domain UMMs achieve strong results on natural images, directly porting them to RS is nontrivial. RS imagery exhibits large-scale variations, complex spatial layouts, dense small objects, and relatively coarse text annotations. These factors make unified RS training more difficult than natural-image settings, especially for models that support both understanding and generation.

Positioning of Our Work: Existing RS vision-language models (RSVLMs) mainly focus on understanding-oriented tasks, while RS image synthesis methods are usually trained in separate generative pipelines. General-domain unified models provide a possible solution, but their direct transfer to RS remains affected by coarse image–text supervision and long-tailed data distributions. In this work, `UniEarth` introduces training and alignment mechanisms tailored to unified RS modeling. Specifically, `UniEarth` introduces intermediate-level cross-modal semantic anchoring via CMA to stabilize generation under coarse supervision, and a DA-UCB strategy via DA-UCB to handle long-tailed distributions. Together, these mechanisms improve joint optimization across both understanding and generation tasks within a single next-token prediction framework.

III. Preliminaries

A. State Space Models

The shared backbone of `UniEarth` is built upon the selective state space model [53] and its improved variant Mamba-2 [54], which bridges the gap between the computational efficiency of CNNs and the long-range dependency modeling of Transformers. Standard SSMs map a 1-D continuous input sequence $x(t) \in \mathbb{R}$ to an output $y(t) \in \mathbb{R}$ through a latent state $h(t) \in \mathbb{R}^N$, governed by a system of linear ordinary differential equations (ODEs)

$$h'(t) = \mathbf{A}h(t) + \mathbf{B}x(t), \quad y(t) = \mathbf{C}h(t) \tag{1}$$

where $\mathbf{A} \in \mathbb{R}^{N\times N}$ is the state transition matrix and $\mathbf{B}, \mathbf{C}$ are projection matrices. We apply zero-order hold (ZOH) discretization with timescale Δ, yielding

$$\overline{\mathbf{A}} = \exp(\Delta\mathbf{A}), \quad \overline{\mathbf{B}} = (\Delta\mathbf{A})^{-1}(\exp(\Delta\mathbf{A}) - \mathbf{I}) \cdot \Delta\mathbf{B}. \tag{2}$$

This discretization enables efficient linear parallel scanning

$$h_t = \overline{\mathbf{A}}h_{t-1} + \overline{\mathbf{B}}x_t, \quad y_t = \mathbf{C}h_t. \tag{3}$$

In `UniEarth`, this mechanism allows high-resolution satellite imagery to be processed with linear complexity $\mathcal{O}(L)$, capturing multiscale geospatial context without the quadratic cost of attention.

B. Multiarmed Bandit

We formulate the dynamic data selection policy as a classical multiarmed bandit problem, where a learner selects one of K arms, indexed by a, at each step n to maximize cumulative reward. The objective is to minimize cumulative regret, defined as the gap between the reward of the optimal arm and that of the selected arm. To reduce regret, UCB [55] assigns an optimistic score to each arm

$$Q_a(n) = \hat{\mu}_a(n) + c \cdot \sqrt{\frac{\ln n}{N_a(n)}}. \tag{4}$$

The empirical mean reward $\hat{\mu}_a(n)$ encourages exploitation by favoring consistently rewarding arms, while the second term promotes exploration by increasing the scores of undersampled arms with small visitation counts $N_a(n)$.

IV. Method

A. Overall Architecture

Fig. 3 shows the overall architecture of `UniEarth`. The model is designed to support both MMU and T2I within a unified AR framework. For MMU, RS images and textual instructions are encoded into a joint sequence, and the model predicts textual responses for tasks such as captioning, visual question answering, and scene classification. For T2I, a text prompt is used as the condition, and the model autoregressively predicts discrete visual tokens, which are then decoded into RS images by the image tokenizer.

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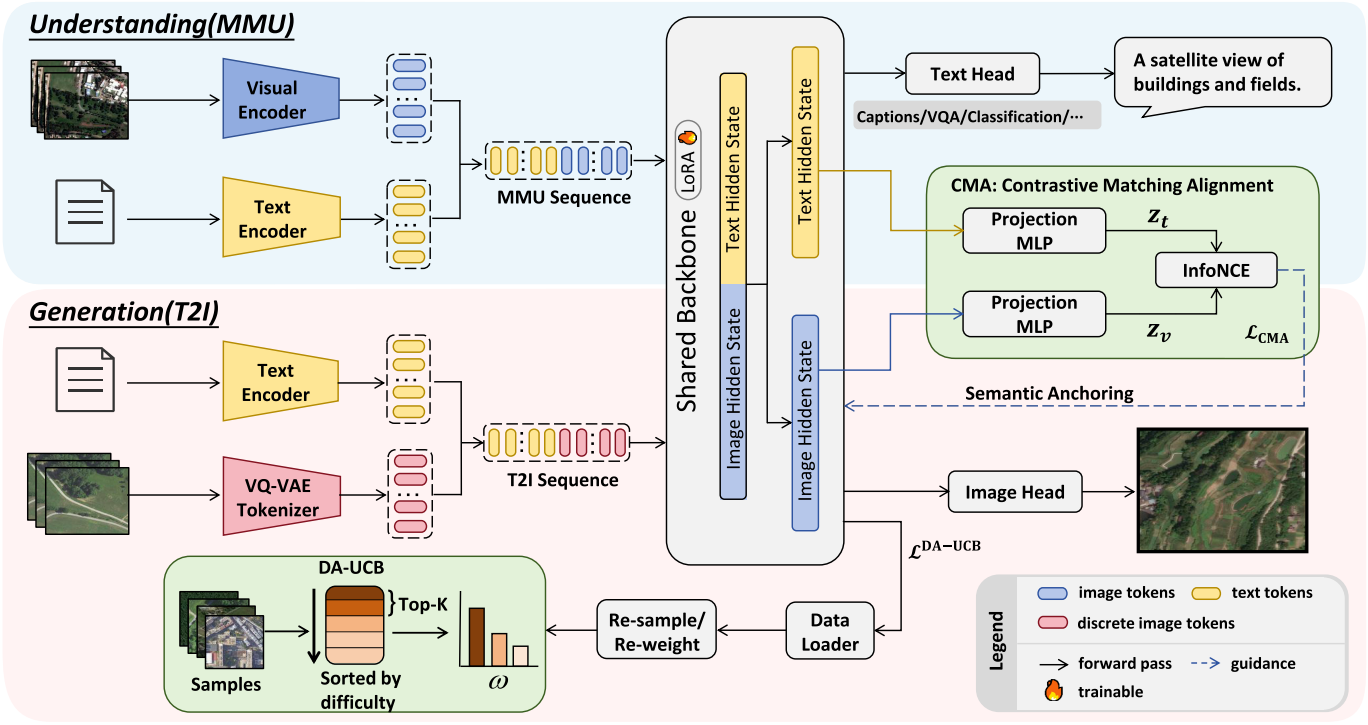


Fig. 3. Overview of UniEarth. The MMU branch predicts textual responses from RS images and instructions, whereas the text-to-image (T2I) branch autoregressively generates discrete visual tokens from text prompts. The two branches share a Mamba-2 backbone with task-specific LoRA modules. CMA aligns their intermediate cross-modal representations, while DA-UCB dynamically reweights samples according to training difficulty.

At the core of UniEarth is a shared Mamba-2 backbone, which provides a competent sequence model for the long visual-token sequences produced by high-resolution RS imagery. By sharing the backbone across MMU and T2I, UniEarth allows the two branches to exchange semantic representations instead of being optimized in isolated pipelines. This shared design enables cross-task knowledge transfer, but it also introduces optimization coupling between MMU and T2I, which is further amplified by the coarse and long-tailed nature of RS supervision.

On top of the shared backbone, UniEarth introduces two domain-specific components to address this coupling. First, CMA imposes cross-modal semantic anchoring on intermediate hidden states, reducing the effect of coarse RS text annotations on generation. Second, DA-UCB adaptively reweights samples according to their dynamic training difficulty, improving optimization under long-tailed RS distributions. Sections IV-B–IV-D describe CMA, DA-UCB, and the overall training objective in detail.

B. Contrastive Matching Alignment

During joint optimization, MMU and T2I interact through the shared backbone, so under-constrained updates from one branch can affect the other. Because RS text supervision is often coarse, scene-level, and semantically ambiguous, the T2I branch is more susceptible to drifting toward texture-level fitting rather than stable semantic grounding. To mitigate this issue, CMA imposes cross-modal semantic anchoring on the intermediate hidden states of the shared backbone. This constraint encourages the T2I-side representations to remain semantically consistent with the input text and supports more stable joint optimization.

We now detail the formulation of CMA. The core idea is to extract modality-specific global representations from the shared backbone's intermediate hidden states and align them in a compact contrastive space, so that matched image–text pairs are pulled closer than mismatched ones. In the following, we describe each step of the pipeline together with the underlying design choice.

Given an image–text pair, let $\mathbf{E}_t \in \mathbb{R}^{L_t \times d}$ and $\mathbf{E}_v \in \mathbb{R}^{L_v \times d}$ denote the input embeddings, where L_t and L_v are the text and visual sequence lengths and d is the hidden dimension. After passing through m Mamba layers, we obtain the intermediate representation $\mathbf{H}^{(m)} \in \mathbb{R}^{(L_t+L_v)\times d}$, which serves as the feature source for cross-modal alignment. We anchor at an intermediate layer rather than at the input embeddings or the final layer for two reasons. Input embeddings are still modality-specific and have not undergone sufficient cross-modal interaction through the selective state-space mechanism, so a contrastive signal there would act only on uncoupled features. Final-layer features, by contrast, are increasingly entangled with the head-side objectives, namely T2I AR prediction and MMU classification, and forcing alignment there would compete with the supervised losses. Intermediate representations therefore provide a more suitable level for shared semantic alignment, since they contain contextualized information while remaining less specialized to a single task. We set the anchoring layer to $m = 3$, corresponding to the third block among the 48 Mamba-2 blocks. This value was selected based on an

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architectural heuristic rather than an exhaustive validation-set search. Specifically, the early intermediate layer allows CMA to operate after initial cross-modal contextualization while avoiding deeper representations that are increasingly specialized toward the T2I and MMU output objectives.

The unified token sequence has a deterministic layout, with text tokens occupying the prefix region and image tokens occupying the subsequent visual-token region. CMA exploits this layout to slice $\mathbf{H}^{(m)}$ into a text-side segment and an image-side segment without introducing any additional separator or learnable aggregation token. Global context vectors are then formed by mean pooling over the token dimension

$$\mathbf{h}_t = \frac{1}{L_t}\sum_{p=1}^{L_t}\mathbf{H}_{t,p}^{(m)}, \qquad \mathbf{h}_v = \frac{1}{L_v}\sum_{q=1}^{L_v}\mathbf{H}_{v,q}^{(m)} \tag{5}$$

where $\mathbf{h}_t, \mathbf{h}_v \in \mathbb{R}^d$ summarize the corresponding text and image segments. Mean pooling is preferred over a CLS-style token because the Mamba-2 backbone has no native aggregation mechanism, and inserting one would require retraining the backbone from scratch. It is also robust to the variable text and visual sequence lengths that are typical in RS data, where caption length and image resolution can both vary across samples.

The pooled vectors are then projected into a shared low-dimensional metric space via modality-specific MLPs

$$\mathbf{z}_t = \mathbf{W}_t^{(2)}\,\mathrm{GELU}\left(\mathbf{W}_t^{(1)}\mathbf{h}_t\right) \tag{6}$$

$$\mathbf{z}_v = \mathbf{W}_v^{(2)}\,\mathrm{GELU}\left(\mathbf{W}_v^{(1)}\mathbf{h}_v\right) \tag{7}$$

where $\mathbf{W}^{(1)} \in \mathbb{R}^{d\times d}$ and $\mathbf{W}^{(2)} \in \mathbb{R}^{d'\times d}$ with $d' \ll d$. This projection serves two purposes. It reduces the dimensionality from d to d', ensuring that the text and image branches are compared in a common compact metric space. At the same time, the use of two separate projection heads, rather than a shared one, allows the network to absorb modality-specific distributional asymmetries inherited from their respective encoders, namely a text encoder and a VQ-VAE tokenizer, before the contrastive comparison. We further apply ℓ_2 normalization to obtain $\hat{\mathbf{z}}_t$ and $\hat{\mathbf{z}}_v$, so that similarity is computed based on angular distance rather than feature magnitude, which keeps InfoNCE well-conditioned in this low-dimensional space.

The pairwise cosine similarity is computed as

$$s_{ij} = \hat{\mathbf{z}}_{t,i}^{\top}\hat{\mathbf{z}}_{v,j} \tag{8}$$

and the bidirectional InfoNCE loss is minimized over a batch of B pairs

$$\mathcal{L}'_{\mathrm{T2I}} = -\frac{1}{B}\sum_{i=1}^{B}\log\frac{\exp(s_{ii}/\tau)}{\sum_{j=1}^{B}\exp(s_{ij}/\tau)} \tag{9}$$

$$\mathcal{L}'_{\mathrm{I2T}} = -\frac{1}{B}\sum_{j=1}^{B}\log\frac{\exp(s_{jj}/\tau)}{\sum_{i=1}^{B}\exp(s_{ij}/\tau)} \tag{10}$$

where τ is the temperature parameter. The two losses contrast matched image–text pairs along the diagonal of the similarity matrix against the mismatched pairs in the same batch. We adopt a contrastive objective rather than direct L2 regression between $\mathbf{z}_t$ and $\mathbf{z}_v$. Under coarse RS text supervision, paired representations may not correspond exactly at the feature level. Directly enforcing point-wise matching may therefore inject noise into the shared backbone. Contrastive alignment only requires matched pairs to be closer than mismatched ones in the embedding space, which better preserves relative semantic structure under ambiguous annotations. This design choice is also empirically supported by the alignment-baseline comparison in Section VI-E2, where CMA consistently outperforms the mse alignment baseline across both generation and understanding metrics. The two directions are complementary: $\mathcal{L}'_{\mathrm{T2I}}$ aligns image features with the text condition for better generation fidelity, while $\mathcal{L}'_{\mathrm{I2T}}$ helps preserve separable text features, since the AR generation loss in (18) supervises the text side only indirectly. The total CMA objective is the average of the two

$$\mathcal{L}_{\mathrm{CMA}} = \frac{1}{2}\left(\mathcal{L}'_{\mathrm{T2I}} + \mathcal{L}'_{\mathrm{I2T}}\right). \tag{11}$$

This mutual anchoring is especially important for RS, where text annotations are coarse-grained. By enforcing bidirectional alignment at the intermediate representation level, CMA encourages both the understanding and generation branches to maintain a semantically coherent and well-grounded shared representation throughout the joint training process. The added cost consists of two modality-specific two-layer MLPs, one for text and one for image, together with one $B \times B$ similarity matrix per batch in the compact d'-dimensional space. The similarity matrix scales quadratically with batch size, but its cost remains negligible compared with the backbone forward pass.

C. Difficulty-Aware Bandit Optimization

RS datasets often exhibit strong long-tailed distributions. Dominant categories such as buildings, vegetation, and roads appear frequently, whereas rare but important concepts such as harbors, bridges, and stadiums are underrepresented. In unified training, this imbalance becomes more severe because the understanding and generation objectives compete for the capacity of the shared backbone. As a result, frequent and easy patterns may dominate gradient updates, while rare or difficult samples are insufficiently explored.

A straightforward solution is to adopt static reweighting or focal-style losses. However, these strategies rely on fixed priors or instantaneous losses, which are not ideal for unified RS training. The difficulty of a sample changes during optimization. A sample that is hard at the early stage may become easy later, while some rare patterns may remain unstable throughout training. Therefore, we propose DA-UCB, which dynamically estimates sample difficulty and allocates training emphasis at the cluster level.

DA-UCB is applied independently to the generation and understanding branches. Let $r \in \{\mathrm{gen}, \mathrm{und}\}$ denote the task branch, where gen denotes T2I and und denotes MMU.

Dynamic Difficulty Estimation: For each sample k in branch r, we maintain a sliding window of recent losses

$$\mathcal{T}_{r,k} = \left[\ell_{r,k}^{(n-M+1)}, \ldots, \ell_{r,k}^{(n)}\right] \tag{12}$$

where M denotes the window size and $\ell_{r,k}^{(n)}$ is the loss of sample k at training step n in branch r. Instead of using only

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the instantaneous loss, we characterize each sample by a 3-D difficulty descriptor

$$\mathbf{f}_{r,k}(n) = \left[\mu_{r,k}(n), \sigma_{r,k}(n), \delta_{r,k}(n)\right] \tag{13}$$

where $\mu_{r,k}(n)$ measures the average difficulty, $\sigma_{r,k}(n)$ reflects the loss volatility, and $\delta_{r,k}(n)$ captures the recent loss trend. The three components are not redundant. The mean $\mu_{r,k}$ summarizes the steady-state difficulty of a sample but is insensitive to learning dynamics. The standard deviation $\sigma_{r,k}$ flags samples whose loss oscillates due to noisy supervision or genuine ambiguity, which respond poorly to fixed reweighting. The trend $\delta_{r,k}$ distinguishes persistently hard samples from samples that are still being learned. The former are candidates for upweighting, whereas the latter can retain the standard training weight. Reweighting based on instantaneous loss alone conflates these three regimes.

Specifically, $\mu_{r,k}(n)$ and $\sigma_{r,k}(n)$ are computed as the mean and standard deviation of the loss trajectory $\mathcal{T}_{r,k}$. To measure whether the sample remains difficult or is being learned, the trend term is defined as

$$\delta_{r,k}(n) = \frac{1}{|\mathcal{T}_{r,k}^{\text{recent}}|} \sum_{\ell \in \mathcal{T}_{r,k}^{\text{recent}}} \ell - \frac{1}{|\mathcal{T}_{r,k}^{\text{old}}|} \sum_{\ell \in \mathcal{T}_{r,k}^{\text{old}}} \ell \tag{14}$$

where $\mathcal{T}_{r,k}^{\text{recent}}$ and $\mathcal{T}_{r,k}^{\text{old}}$ denote the recent and earlier halves of the loss window, respectively. A positive trend indicates that the sample has not been sufficiently learned or becomes more difficult, while a negative trend suggests that the sample is being gradually fit.

Cluster-Level Bandit Policy: We deliberately operate at the cluster level rather than the sample level. With $|\mathcal{D}|$ on the order of 10^6, treating each sample as a separate arm a would produce visitation counts $N_a(n)$ that grow extremely slowly relative to the total step count n. Consequently, the exploration term $(\ln n/N_a(n))^{1/2}$ may dominate the empirical reward $\hat{\mu}_a(n)$. This imbalance can cause the policy to degenerate into near-uniform sampling. Clustering compresses the arm space from millions of sample-level arms to K_c cluster-level arms, so visitation statistics accumulate fast enough for the exploitation term to remain informative.

Every T_{cluster} iterations, samples are grouped into K_c difficulty clusters using K-Means over $\mathbf{f}_{r,k}(n)$. Before clustering, each dimension of $\mathbf{f}_{r,k}(n)$ is standardized to zero mean and unit variance within the tracked sample pool, preventing one component from dominating due to scale differences. Each difficulty cluster is regarded as an arm in a multiarmed bandit problem. Let $\mathcal{C}_{r,j}$ denote the jth difficulty cluster in branch r, and let $a_r(k)$ denote the cluster assignment of sample k. Equation (15) instantiates the generic UCB formulation in (4) independently for each branch

$$Q_{r,j}(n) = \bar{\mu}_{r,j}(n) + c\sqrt{\frac{\ln n}{N_{r,j}(n)}}. \tag{15}$$

Here, the generic arm index a in (4) corresponds to the jth difficulty cluster $\mathcal{C}_{r,j}$ in branch r. Accordingly, $\hat{\mu}_a(n)$ and $N_a(n)$ are instantiated as $\bar{\mu}_{r,j}(n)$ and $N_{r,j}(n)$, respectively. The quantity $\bar{\mu}_{r,j}(n)$ is the mean difficulty of the cluster and serves as its empirical reward, while c controls the strength of exploration. The first term favors clusters that remain difficult, while the second term encourages the model to revisit under-sampled clusters. This design prevents the model from over-concentrating on frequent and easy patterns.

Since K-Means is performed repeatedly during training, the identity of a cluster may change across reclustering rounds. To maintain consistent visitation counts, after each reclustering step, the newly obtained centroids are matched to the previous centroids according to nearest-neighbor distance. The visitation counts are then inherited based on the matched cluster identities, while newly appeared clusters are initialized with a visitation count of one.

Weighted Joint Optimization: The clusters with the highest UCB scores are selected as hard arms

$$\mathcal{A}_{r,\text{hard}} = \text{TopK}_{j\in\{1,\ldots,K_c\}} Q_{r,j}(n) \tag{16}$$

where TopK selects the K_{hard} clusters with the largest UCB scores among the K_c candidate clusters, and $\mathcal{A}_{r,\text{hard}}$ denotes their index set in branch r. The sample weight is then defined as

$$w_{r,k}^{(n)} = \begin{cases} \omega_{\text{hard}}, & a_r(k) \in \mathcal{A}_{r,\text{hard}} \\ 1.0, & \text{otherwise} \end{cases} \tag{17}$$

where ω_{hard} is the weight assigned to samples from hard clusters. Before the first clustering step, all samples are assigned unit weights.

Applying DA-UCB independently to the two branches is important because the generation and understanding tasks have different loss scales, output lengths, and convergence behaviors. Branch-specific weighting avoids allowing one task to dominate the shared optimization, while still enabling both branches to benefit from difficulty-aware training.

Algorithm 1 DA-UCB

Require: Window size M, cluster number K_c, clustering period T_{cluster}, hard-arm number K_{hard}, hard-arm weight ω_{hard}, exploration coefficient c

Initialize $\mathcal{T}_{r,k} \leftarrow \emptyset$, $N_{r,j} \leftarrow 1$, $\mathcal{A}_{r,\text{hard}} \leftarrow \emptyset$, $\mathcal{M}_r \leftarrow \emptyset$
for training step $n = 1, 2, \ldots$ **do**
  for branch $r \in \{\text{gen}, \text{und}\}$ **do**
    Difficulty estimation:
    Append $\ell_{r,k}^{(n)}$ to $\mathcal{T}_{r,k}$
    $\mathbf{f}_{r,k}(n) \leftarrow [\mu_{r,k}, \sigma_{r,k}, \delta_{r,k}]$ via Eq. (13)
    if n mod $T_{\text{cluster}} = 0$ **then**
      Bandit update:
      $\{\mathcal{C}_{r,j}\} \leftarrow \text{KMEANS}(\mathbf{f}_{r,k}(n))$
      $N_{r,j} \leftarrow \text{MATCH}(\mathcal{C}_{r,j}, \mathcal{M}_r)$
      $\mathcal{A}_{r,\text{hard}} \leftarrow \text{TOPK}_{K_{\text{hard}}}\, Q_{r,j}(n)$ via Eq. (16)
      $\mathcal{M}_r \leftarrow \{\mathcal{C}_{r,j}\}$, $N_{r,j}$ += $\mathbb{1}[j \in \mathcal{A}_{r,\text{hard}}]$
    end if
    Reweighting:
    $w_{r,k}^{(n)} \leftarrow \omega_{\text{hard}}$ if $a_r(k) \in \mathcal{A}_{r,\text{hard}}$, else 1
    $\mathcal{L}_r^{\text{DA-UCB}} \leftarrow \sum_b w_r^{(b)} \mathcal{L}_r^{(b)} \big/ \sum_b w_r^{(b)}$
  end for
end for

Unlike focal loss, which emphasizes high-loss samples at the instance level, DA-UCB performs difficulty-aware allocation at the cluster level. This makes the training signal more

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stable and reduces the sensitivity to noisy individual losses. Unlike class-balanced reweighting, DA-UCB does not require explicit category-frequency annotations and can be applied to captioning, VQA, classification, and generation within the same unified framework. The overall procedure is summarized in Algorithm 1.

D. Overall Training Objective of UniEarth

We first define the per-branch weighted losses produced by DA-UCB. At each training step, the weights $w_{r,k}^{(n)}$ from (17) are applied to the per-sample losses to form the weighted branch losses

$$\mathcal{L}_{\text{gen}}^{\text{DA-UCB}} = \frac{\sum_{b=1}^{B} w_{\text{gen}}^{(b)} \mathcal{L}_{\text{gen}}^{(b)}}{\sum_{b=1}^{B} w_{\text{gen}}^{(b)}} \tag{18}$$

$$\mathcal{L}_{\text{und}}^{\text{DA-UCB}} = \frac{\sum_{b=1}^{B} w_{\text{und}}^{(b)} \mathcal{L}_{\text{und}}^{(b)}}{\sum_{b=1}^{B} w_{\text{und}}^{(b)}} \tag{19}$$

where $\mathcal{L}_{\text{gen}}^{(b)}$ is the per-sample AR token prediction loss for T2I, and $\mathcal{L}_{\text{und}}^{(b)}$ is the per-sample cross-entropy loss for MMU.

We then incorporate CMA into the T2I generation branch. The reformulated branch objective is

$$\mathcal{L}_{\text{T2I}} = \mathcal{L}_{\text{gen}}^{\text{DA-UCB}} + \lambda \mathcal{L}_{\text{CMA}} \tag{20}$$

where λ is the alignment weight controlling the strength of CMA. The CMA term is computed at the batch level over all paired samples and is not subject to DA-UCB reweighting, since it serves as a global semantic-anchoring regularizer rather than a per-sample supervision signal.

The overall training objective combines the MMU loss and the T2I generation loss

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{und}}^{\text{DA-UCB}} + \mathcal{L}_{\text{T2I}} = \mathcal{L}_{\text{und}}^{\text{DA-UCB}} + \mathcal{L}_{\text{gen}}^{\text{DA-UCB}} + \lambda \mathcal{L}_{\text{CMA}}. \tag{21}$$

The above steps are iterated until UniEarth converges or reaches the maximum number of training rounds.

V. Training Data

We train UniEarth on a corpus of approximately 1-M RS samples, drawn from Git-10M [12] for pretraining and from established RS benchmarks for fine-tuning. The corpus spans four downstream task families, namely scene classification, image captioning, visual question answering, and T2I, all reformulated under a single next-token prediction interface. Detailed statistics are summarized in Table I.

A. Pretraining Data

For pretraining, we source image–text pairs from Git-10M, one of the largest publicly available RS image–text corpora. Git-10M is collected at a global scale and covers diverse geographic regions and scene types, including urban, agricultural, coastal, and mountainous environments. Each image is paired with a textual caption generated through a combination of rule-based templates and large vision-language models, which provides scalable but heterogeneous supervision compared with manually annotated datasets.

We sample two subsets from Git-10M for the two branches in UniEarth. For the MMU branch, we sample 676-K image–text pairs covering diverse scene-level semantics, including urban areas, agricultural fields, water bodies, forests, and transportation infrastructure, to provide broad visual-language coverage for cross-modal representation learning. For the T2I generation branch, we sample 83-K image–text pairs from the same source. The two subsets are drawn from a shared pool, and no benchmark-specific or quality-based selection is applied beyond format and length filtering used to remove malformed entries. The two subsets differ in size because the two branches have different supervision requirements. The MMU branch benefits from broader scene-level coverage for learning cross-modal alignment. In contrast, the T2I branch performs dense image-token generation, for which additional samples yield diminishing returns under the same compute budget.

Before training, all images are resized to a uniform input resolution while captions are tokenized using the text tokenizer of the shared backbone. For the T2I branch, images are tokenized into discrete visual tokens using a VQ-VAE. Sample-level metadata available in Git-10M, such as image source and acquisition information, is not used during training, since UniEarth relies only on raw image and caption pairs as supervision.

B. Fine-Tuning Data

For fine-tuning, we collect approximately 430 K annotated samples spanning four downstream tasks, drawn from established RS benchmarks to ensure task diversity and evaluation fairness. Each task is associated with one or more datasets that provide coverage in scene type and annotation style.

Scene classification combines AID [59] and NWPU-RESISC45 [60] for broad scene coverage. AID provides aerial imagery across 30 categories collected from Google Earth at varying spatial resolutions, while NWPU-RESISC45 covers 45 scene categories at resolutions from 0.2 to 30 m per pixel. Image captioning combines RSICD [56], UCM-Captions [57], Sydney-Captions [57], and a subset of Git-10M. The first three datasets contribute curated captions over established scene-category benchmarks, and the Git-10M subset adds large-scale image–text pairs sampled from the same pretraining source. Visual question answering combines RSVQA-LR [58] for presence and comparison questions with a subset of RSIVQA [35] to cover more diverse question types beyond presence and comparison. T2I fine-tuning uses the RSICD [56] training split with image–text pairs across 30 scene categories. Per-dataset statistics are reported in Table I.

These four tasks differ substantially in supervision density, output length, and semantic granularity. Scene classification and VQA produce short textual outputs anchored on a small label space, whereas captioning and T2I require richer semantic modeling and longer AR sequences. We organize all of them within a single next-token prediction framework, with task-specific instruction templates used to distinguish different tasks at input time. For instance, classification samples are reformatted with prompts that ask for the scene category, while captioning samples use prompts that request a description of the image. The same backbone parameters then handle every task, and the only task-conditioning signal at training

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TABLE I
DETAILED STATISTICS OF THE SOURCE DATASETS

Task	Dataset	Source Dataset Size	Image Size
Image Captioning	RSICD [56]	54,605	224×224
	UCM-Captions [57]	10,500	256×256
	Sydney-Captions [57]	3,065	500×500
	Git-10M [12] (sampled)	10,000,000	–
Visual Question Answering	RSVQA-LR [58]	77,232	256×256
	RSIVQA [35] (sampled)	111,134	–
Scene Classification	AID [59]	10,000	600×600
	NWPU-RESISC45 [60]	31,500	256×256
Text-to-Image Generation	Git-10M [12] (HQ subset)	10,000,000	–
	RSICD [56]	54,605	224×224

Note: "–" indicates variable image sizes in the source datasets.

time is the instruction prefix. This unified format exposes the shared backbone to heterogeneous supervision signals within a single training loop, providing a common interface for learning transferable RS representations.

C. Data Decontamination

To prevent data leakage and ensure fair evaluation, we apply a strict decontamination procedure to the pretraining corpus before any model training begins. We remove from the Git-10M pretraining subset all samples that overlap with the test splits of any evaluation benchmark. Overlap is determined by perceptual image hashing (pHash), which is robust to minor resizing and compression artifacts that may exist across different dataset versions. Concretely, for every Git-10M candidate, we compute its perceptual hash and compare it against the hashes of all test images, rejecting candidates whose hash matches a test image up to compression and resizing tolerance. We verify that no test image from RSICD, UCM-Captions, RSVQA-LR, AID, or NWPU-RESISC45 appears in our pretraining or fine-tuning corpus. RSIVQA is used only as auxiliary fine-tuning data and not as an evaluation benchmark, so it is not included in this verification. This procedure ensures that all reported results reflect genuine generalization rather than memorization of training data.

VI. EXPERIMENTS

To extensively validate the effectiveness of `UniEarth`, we conduct evaluations on five standard benchmarks covering both MMU, namely RSVQA-LR [58], UCM-Captions [57], AID [59], and NWPU-RESISC45 [60], and T2I, namely RSICD [56]. We compare `UniEarth` against a diverse set of advanced baselines, including task-specific specialist models, general-purpose VLMs, and domain-specific RSVLMs. The experimental section is organized to follow the unified design of `UniEarth`: Section VI-B reports MMU results across captioning, visual question answering, and scene classification; Section VI-C reports T2I results; and Sections VI-D–VI-F present ablation studies and mechanism-level analyses on the CMA module and the DA-UCB strategy.

A. Implementation Details

`UniEarth` is trained with a two-stage strategy. In Stage 1, the MMU and T2I branches are pretrained independently. In Stage 2, all LoRA modules are fine-tuned within the shared backbone for cross-task interaction and unified optimization.

For the shared backbone, we initialize from the publicly available OmniMamba checkpoint and adapt it to the RS domain. The shared backbone has $n_{\text{layer}} = 48$ Mamba-2 blocks and a hidden dimension of $d_{\text{model}} = 2048$. All domain-specific components, including CMA projectors, DA-UCB scheduling logic, and task-specific LoRA modules, are trained from scratch. For the MMU branch, we adopt a dual-stream visual encoder that integrates DINOv2 [61] and SigLIP [62] to capture both fine-grained and semantic visual features. For T2I generation, we use a VQ-VAE from LlamaGen [63], further adapted to RS imagery to produce discrete visual tokens.

Stage 1 (Independent Pretraining): In this stage, the MMU and T2I branches are trained separately, with only task-specific LoRA modules and projectors activated while the shared backbone remains frozen. Stage 1 experiments are conducted on eight NVIDIA RTX 4090 GPUs. For the MMU branch, we use a learning rate of 1×10^{-3} and a batch size of 2 with gradient accumulation. For the T2I branch, we use a learning rate of 2×10^{-4} and a batch size of 10 with gradient accumulation. Both branches use the AdamW optimizer with $\beta_1 = 0.9$ and $\beta_2 = 0.95$, together with a cosine learning-rate decay schedule.

Stage 2 (Unified Fine-Tuning): In this stage, all LoRA modules are jointly fine-tuned within the shared backbone. Stage 2 experiments are conducted on eight NVIDIA A100 GPUs with 80-GB memory using `bf16` mixed precision. We use the AdamW optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.95$, weight decay of 0.01, and gradient clipping with a maximum norm of 1.0. The initial learning rate is set to 5×10^{-5} and decayed with a cosine schedule, where the minimum learning-rate ratio is 0.1. At each optimization step, we use batch sizes of 42 and 3 for the T2I and MMU branches, respectively. This asymmetric setting was chosen to account for the different computational costs and training throughput of the two branches while maintaining stable joint training. For DA-UCB, we set the loss window size $M = 50$, the cluster number $K_c = 8$, the exploration coefficient $c = 2.0$, the hard-cluster weight $\omega_{\text{hard}} = 2.0$, and the number of selected hard clusters $K_{\text{hard}} = 3$. Cluster centroids are periodically refreshed during training, and DA-UCB reweighting is activated after an

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initial warm-up phase, during which all samples are assigned unit weights. For CMA, the alignment weight λ is set to 0.5 by default.

B. Multimodal Understanding

We evaluate the MMU capability of UniEarth on three representative tasks: image captioning, visual question answering, and scene classification. Together, these tasks cover sentence-level generation, short-answer reasoning, and label-level recognition over RS imagery.

1) Image Captioning:

a) Dataset and metric: UCM-Captions [57] is derived from the UC Merced Land-Use dataset, consisting of 2100 RS images across 21 land-use categories with a spatial resolution of approximately 0.3 m per pixel. Each image is annotated with five reference captions. Following standard protocols [64], [65], we use 80% of the data for training, 10% for validation, and 10% for testing.

We report four standard metrics for image captioning evaluation. BLEU-4 evaluates precision by counting matching 4-g between generated and reference captions. METEOR considers both precision and recall at the unigram level, incorporating stemming and synonym matching for more flexible evaluation. ROUGE-L measures the longest common subsequence between generated and reference sentences, capturing sentence-level structural similarity. CIDEr is specifically designed for image captioning evaluation, measuring semantic consensus by weighting n-grams based on TF-IDF scores computed across the reference corpus.

b) Result: Table II reports results on the UCM-Captions dataset. We compare UniEarth against understanding-only specialist models, general-domain VLMs, RS-specific VLMs, and unified models.

Among understanding-only specialists, SAA [64] and post-processing [65] achieve CIDEr scores of 294.51 and 309.64, respectively. General-domain VLMs under zero-shot settings perform poorly on this task. MiniGPT-4 [66] obtains a CIDEr of only 0.03, and DeepSeek-VL2 [69] reaches 26.42, indicating that these models lack the domain knowledge required for accurate RS scene description. Among RSVLMs, RingMo-Agent [30] achieves the previous best CIDEr of 373.68. RS-LLaVA and RS-CapRet [73] also show competitive BLEU-4 scores of 72.84 and 67.00, respectively.

UniEarth achieves the best performance across all four metrics, with BLEU-4 of 88.12, METEOR of 59.48, ROUGE-L of 92.40, and CIDEr of 448.94. Compared with RingMo-Agent, the previous best RSVLM on CIDEr, UniEarth improves CIDEr by 75.26 points, a 20.1% relative gain, and improves BLEU-4 and ROUGE-L by 10.49 and 6.89 points, respectively. Compared with OmniMamba FT trained on the same RS corpus, UniEarth still obtains clear gains across all metrics, including a 97.19-point improvement in CIDEr. These results show that UniEarth provides stronger RS captioning performance within the unified understanding-generation framework. The CIDEr improvement further indicates better agreement with reference captions in RS scene descriptions.

2) Visual Question Answering:

a) Dataset and metric: RSVQA-LR [58] is a large-scale visual question answering benchmark for low-resolution RS imagery. It contains 772 images at 256×256 pixels derived from Sentinel-2 satellite data, with a total of 77 232 question-answer pairs spanning four question types, namely presence, comparison, counting, and rural/urban classification, following the official RSVQA-LR protocol. We follow the official train/test split for evaluation. ZS and FT denote zero-shot and fine-tuning settings, respectively. Pre. Acc denotes the accuracy on presence questions, while Comp. Acc denotes the accuracy on comparison questions. Avg. Acc (P&C) is the average accuracy over presence and comparison questions, whereas Avg. Acc (All) is the overall accuracy over all four RSVQA-LR question types: presence, comparison, counting, and rural/urban classification.

We adopt Top-1 accuracy as the primary evaluation metric. An answer is considered correct if the model's generated text exactly matches the ground-truth answer.

b) Result: Table III summarizes results on the RSVQA-LR dataset. We compare UniEarth against specialist VQA models, general-domain VLMs, RSVLMs, and unified models.

Among specialist models, Bi-Modal [33] and SHRNet [34] achieve Avg. Acc (P&C) of 91.11% and 90.76%, respectively, representing strong task-specific baselines. General-domain VLMs under zero-shot settings perform significantly worse. The best-performing Qwen3-VL [70] only reaches 72.01% Avg. Acc (P&C), whereas the RS-specific models generally reach around 90% or higher. This gap highlights the limited zero-shot transferability of general-purpose VLMs to RS VQA tasks.

Among RSVLMs, RingMo-Agent [30] attains the previous best presence accuracy of 93.10%, RSGPT [10] achieves the highest previous comparison accuracy of 91.70%, and RS-LLaVA [26] reaches the best previous Avg. Acc (P&C) of 92.07%. Under the unified model category, OmniMamba [18] FT achieves 92.50% on presence questions but only 86.13% on comparison questions, yielding an Avg. Acc (All) of 83.81%.

UniEarth attains 97.36% accuracy on presence questions and 97.51% accuracy on comparison questions, surpassing RingMo-Agent by 4.26 percentage points in presence accuracy and RSGPT by 5.81 percentage points in comparison accuracy. UniEarth further achieves an Avg. Acc (P&C) of 97.44% and an Avg. Acc (All) of 91.84%, both state-of-the-art. The strong performance on comparison questions is particularly noteworthy, as these questions require relational reasoning between multiple object categories, demonstrating that UniEarth's shared representations effectively support complex semantic understanding. Notably, UniEarth outperforms all RS-specific fine-tuned models despite operating as a unified generation-understanding framework rather than a VQA-specialized model, validating the benefit of joint optimization with CMA and DA-UCB.

3) Scene Classification:

a) Dataset and metric: We evaluate scene classification on two widely used benchmarks. AID [59] contains 10 000 aerial images across 30 scene categories collected from Google

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TABLE II
QUANTITATIVE RESULTS OF UNIEARTH FOR MMU ON THE UCM-CAPTIONS DATASET

Type	Model	Train	BLEU-4↑	METEOR↑	ROUGE-L↑	CIDEr↑
Understanding-only Models						
Specialist	SAA [64]	FT	64.77	38.59	69.42	294.51
	Post-processing [65]	FT	62.62	40.80	74.06	309.64
VLMs	MiniGPT-4 [66]	ZS	18.10	33.36	41.37	0.03
	Shikra [67]	ZS	33.98	32.56	56.73	56.69
	MiniGPT-v2 [68]	ZS	36.16	32.41	56.57	60.66
	DeepSeek-VL2 [69]	ZS	5.44	15.95	27.22	26.42
	Qwen3-VL [70]	ZS	1.82	15.73	24.92	8.85
	Gemini 2.5 [71]	ZS	2.78	17.05	25.96	11.74
	GPT-4o [72]	ZS	3.33	15.37	26.31	21.13
RSVLMs	RSGPT [10]	FT	65.74	42.21	78.34	333.23
	RingMoGPT [29]	FT	–	49.90	83.29	359.32
	RS-CapRet [73]	FT	67.00	47.20	81.70	354.80
	RS-LLaVA [26]	FT	72.84	47.98	85.17	349.43
	SkyEyeGPT [28]	FT	<u>78.41</u>	46.24	79.49	236.75
	RingMo-Agent [30]	FT	77.63	<u>51.79</u>	<u>85.51</u>	<u>373.68</u>
Unified Models						
VLMs	Janus-Pro [15]	ZS	2.50	16.05	19.38	0.30
	Show-o2 [74]	ZS	0.88	11.42	11.38	1.15
	OmniMamba [18]	ZS	6.55	6.60	16.43	1.27
	OmniMamba [18]	FT	72.67	49.57	81.58	351.75
RSVLMs	**UniEarth (Ours)**	**FT**	**88.12**	**59.48**	**92.40**	**448.94**

The best results are highlighted in **bold**, and the second-best are <u>underlined</u>. FT indicates fine-tuning, while ZS indicates zero-shot evaluation.

Earth at varying spatial resolutions, with an image size of 600 × 600 pixels. NWPU-RESISC45 [60] is a larger and more challenging dataset, comprising 31 500 RS images across 45 scene categories with 700 images per category. The image size is 256 × 256 pixels, and the spatial resolution ranges from 0.2 to 30 m per pixel. For both datasets, we follow the standard evaluation protocol with a 20% training ratio, namely TR = 20%.

We adopt Top-1 classification accuracy as the evaluation metric. UniEarth performs classification by generating the scene category name as text. An output is considered correct if the generated text contains the ground-truth category label.

b) Result: Table IV reports the classification accuracy on AID and NWPU-RESISC45. Among understanding-only specialists, LSENet [79] achieves 94.41% on AID and 93.34% on NWPU, while SeCo-ResNet-50 [80] obtains 93.47% and 92.91%, respectively. General-domain VLMs under zero-shot settings show highly variable performance. Among the evaluated general-purpose VLMs, Qwen3-VL [70] achieves the strongest overall classification performance, reaching 70.71% on AID and 71.01% on NWPU-RESISC45, while other models, including DeepSeek-VL2 [69], Gemini 2.5 [71], and GPT-4o [72], exhibit more variable performance across the two benchmarks. These results indicate that a substantial gap remains between general-purpose VLMs and RS-oriented models for scene recognition. OmniMamba [18] without fine-tuning also performs poorly, achieving only 2.66% and 2.02% on AID and NWPU-RESISC45, suggesting that its visual tokenization does not generalize to RS scene categories.

Among RSVLMs, RingMo-Agent [30] achieves 92.43% on AID and 93.96% on NWPU, while EarthGPT [6] reaches 93.84% on NWPU. Under the unified model category, OmniMamba FT attains 90.61% on AID and 89.38% on NWPU after fine-tuning on our RS corpus.

UniEarth achieves 96.11% on AID and 95.73% on NWPU, outperforming all baselines on both benchmarks. Compared with OmniMamba FT, UniEarth improves accuracy by 5.50 percentage points on AID and 6.35 percentage points on NWPU, showing that the gain is not solely from domain-matched fine-tuning. Compared with the best specialist model LSENet, UniEarth also achieves a clear margin of +1.70 percentage points on AID and +2.39 percentage points on NWPU. These results indicate that the unified understanding-generation framework also benefits scene-level recognition.

4) Qualitative Results on Understanding: We further illustrate the MMU ability of UniEarth qualitatively, as shown in Fig. 4. For complex RS scenes containing multiple structural elements, UniEarth produces scene descriptions that not only identify the dominant scene category but also capture finer details such as the presence of running tracks, parking lots, road boundaries, and surrounding buildings. This suggests that the shared representation learned by UniEarth preserves both global scene context and local structural cues, which are essential for RS image interpretation. The descriptions remain consistent with the spatial layout of the input image, indicating that the model can ground textual outputs to specific structural patterns rather than relying solely on dominant scene-level priors.

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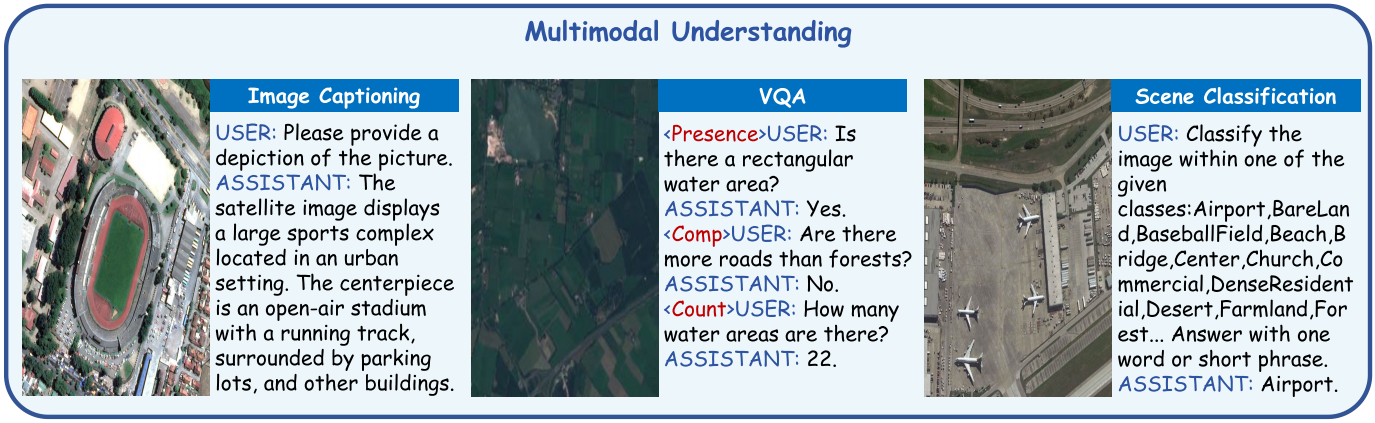


Fig. 4. Qualitative results of UniEarth on MMU tasks, including image captioning, visual question answering, and scene classification.

TABLE III
QUANTITATIVE RESULTS OF UNIEARTH FOR MMU ON THE RSVQA-LR

Model	Train	Pre. Acc↑	Comp. Acc↑	Avg. Acc↑ P&C	Avg. Acc↑ All
Understanding-only Models					
Specialist models					
EasyToHard [32]	FT	90.66	87.49	89.08	–
Bi-Modal [33]	FT	91.06	91.16	91.11	–
SHRNet [34]	FT	91.03	90.48	90.76	–
VLMs					
MiniGPT-4 [66]	ZS	43.86	57.55	50.71	–
Shikra [67]	ZS	46.47	60.31	53.39	–
Qwen-VL [75]	ZS	38.57	67.59	53.08	–
MiniGPT-v2 [68]	ZS	49.85	63.09	56.47	–
InstructBLIP [76]	ZS	48.83	65.92	57.38	–
mPLUG-Owl2 [77]	ZS	74.04	63.69	68.87	–
LLaVA-1.5 [78]	ZS	55.46	68.20	61.83	–
Qwen3-VL [70]	ZS	66.53	77.49	72.01	–
Gemini 2.5 [71]	ZS	67.99	72.77	70.38	–
GPT-4o [72]	ZS	69.20	63.14	66.17	–
RSVLMs					
RSGPT [10]	FT	91.17	91.70	91.44	–
GeoChat [8]	FT	91.09	90.33	90.71	–
LHRS-Bot [31]	FT	89.07	88.51	88.79	–
H^2RSVLM [27]	FT	89.58	89.79	89.69	–
RS-LLaVA [26]	FT	92.80	91.33	92.07	–
SkyEyeGPT [28]	FT	88.93	88.63	88.78	–
RingMo-Agent [30]	FT	93.10	87.50	90.30	–
Unified Models					
VLMs					
Janus-Pro [15]	ZS	64.81	52.37	58.59	41.21
Show-o2 [74]	ZS	53.33	28.21	40.77	32.69
OmniMamba [18]	ZS	67.95	58.58	63.27	51.10
OmniMamba [18]	FT	92.50	86.13	89.32	83.81
RSVLMs					
UniEarth (Ours)	**FT**	**97.36**	**97.51**	**97.44**	**91.84**

[1]The best results are highlighted in **bold**, and the second-best are underlined. FT indicates fine-tuning, while ZS indicates zero-shot evaluation. Pre. Acc denotes the accuracy on presence questions, while Comp. Acc denotes the accuracy on comparison questions.

[2]Avg. Acc (P&C) is the average accuracy over presence and comparison questions. Avg. Acc (All) follows the official RSVQA-LR protocol over all four question types (presence, comparison, counting, and rural/urban classification); "–" indicates that the corresponding method does not report results on the additional question types.

TABLE IV
QUANTITATIVE RESULTS OF UNIEARTH FOR MMU ON AID AND NWPU DATASETS

Model	Train	Acc (%)↑ AID	Acc (%)↑ NWPU
Understanding-only Models			
Specialist models			
LSENet [79]	FT	94.41	93.34
SeCo-ResNet-50 [80]	FT	93.47	92.91
VLMs			
DeepSeek-VL2 [69]	ZS	35.63	17.60
Qwen3-VL [70]	ZS	70.71	71.01
Gemini 2.5 [71]	ZS	37.97	68.58
GPT-4o [72]	ZS	58.33	50.52
RSVLMs			
EarthGPT [6]	FT	–	93.84
RingMo-Agent [30]	FT	92.43	93.96
Unified Models			
VLMs			
Janus-Pro [15]	ZS	59.49	54.75
Show-o2 [74]	ZS	63.61	74.51
OmniMamba [18]	ZS	2.66	2.02
OmniMamba [18]	FT	90.61	89.38
RSVLMs			
UniEarth (Ours)	**FT**	**96.11**	**95.73**

The best results for each dataset are highlighted in **bold**, and the second-best are underlined. FT indicates fine-tuning, while ZS indicates zero-shot evaluation.

C. Image Generation

We next evaluate the T2I capability of UniEarth. This task complements the MMU tasks above and evaluates whether the shared backbone can support high-fidelity RS synthesis under coarse text supervision.

1) T2I Generation:

a) Dataset and metric: RSICD [56] is a standard benchmark for RS image captioning and generation. It contains 10 921 RS images covering 30 scene categories at a resolution of 224 × 224 pixels. Each image is annotated with five natural language descriptions. Following prior work [12], we use the standard test split for T2I evaluation.

We evaluate generation performance using both visual fidelity and semantic alignment metrics. For quality

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TABLE V
QUANTITATIVE RESULTS OF UNIEARTH FOR IMAGE GENERATION ON THE RSICD DATASET

Method	Train	FID↓	CLIP Score↑
Generation-only Models			
General generative models			
Attn-GAN [83]	FT	95.81	20.19
DAE-GAN [37]	FT	93.15	19.69
DF-GAN [38]	FT	109.41	19.76
Lafite [40]	FT	74.11	22.52
DALL-E [84]	FT	191.93	20.13
RS generative models			
Txt2Img-MHN$_{vqvae}$ [39]	FT	175.36	21.35
Txt2Img-MHN$_{vqgan}$ [39]	FT	102.44	20.27
RSDiff [11]	FT	66.49	–
CRS-Diff [13]	FT	50.72	20.33
Text2Earth [12]	FT	<u>24.49</u>	25.62
Unified Models			
VLMs			
Janus-Pro [15]	ZS	310.40	19.81
Show-o2 [74]	ZS	158.50	<u>29.71</u>
OmniMamba [18]	ZS	167.71	18.64
OmniMamba [18]	FT	45.80	19.81
RSVLMs			
UniEarth (Ours)	**FT**	**13.56**	**30.51**

The best results for each dataset are highlighted in **bold**, and the second-best are <u>underlined</u>. FT indicates fine-tuning, while ZS indicates zero-shot evaluation.

assessment, we employ the FID [81], which measures the distribution distance between generated and real images using features extracted from the Inception-v3 network. A lower FID indicates better distributional similarity. For semantic accuracy, we use the CLIP Score [82], which measures the cosine similarity between the CLIP embeddings of generated images and their text prompts. A higher CLIP Score indicates better image–text alignment.

b) Result: Table V presents the T2I results on RSICD. Baselines are organized into four groups: general generation models, RS generation models, general-domain unified VLMs, and unified RSVLMs.

Among general generation models, Attn-GAN [83], DAE-GAN [37], and DF-GAN [38] yield FID scores above 90, and DALL-E [84] reaches 191.93, reflecting the difficulty of adapting general-domain architectures to RS imagery. Lafite [40] achieves a lower FID of 74.11 through language-free training.

Among RS generation models, Txt2Img-MHN [39] and RSDiff [11] improve over the GAN-based methods. CRS-Diff [13] achieves an FID of 50.72, and the current best specialist, Text2Earth, reaches 24.49. However, these models are designed exclusively for generation and cannot perform understanding tasks.

Among unified VLMs, models such as Janus-Pro [15], Show-o [16], and OmniMamba [18] exhibit substantially higher FID scores from 158 to 310 under zero-shot settings, indicating that general-domain unified architectures struggle with RS imagery without domain adaptation. Even after fine-tuning on our RS corpus, OmniMamba FT only achieves an FID of 45.80, still far from specialist-level performance.

TABLE VI
ABLATION ON THE EFFECTIVENESS OF DA-UCB AND CMA

Setting		RSICD		RSVQA-LR	UCM-Captions	
DA-UCB	CMA	FID↓	CLIP Score↑	Avg. Acc (All)↑	BLEU-4↑	CIDEr↑
×	×	45.80	19.81	83.81	72.67	351.75
×	✓	15.75	23.01	83.87	80.10	392.84
✓	×	19.47	23.22	86.08	82.84	420.81
✓	✓	**13.56**	**30.51**	**91.84**	**88.12**	**448.94**

UniEarth achieves a state-of-the-art FID of 13.56 and a CLIP Score of 30.51, significantly outperforming Text2Earth by 44.6% in FID and surpassing it in CLIP Score by 4.89 points. Compared with OmniMamba FT, UniEarth reduces FID from 45.80 to 13.56, corresponding to a 70.4% relative reduction, while improving CLIP Score by 10.70 points. This shows that the improvement is not solely due to RS-domain fine-tuning. These results indicate that UniEarth substantially narrows the gap between unified models and specialist RS generation models. The higher CLIP Score further suggests improved image–text alignment under coarse RS text supervision.

2) Qualitative Results on Generation: Fig. 5 illustrates the image generation behavior of UniEarth. The model synthesizes satellite-like images that are visually plausible and well aligned with the textual prompts, including building rooftops beside shaded streets, roads along forested areas, and dense suburban housing. It also captures scene-specific foreground structures and land-cover compositions, such as circular oil tanks, forest edges, and exposed riverbank sandbars. Compared with the failure modes observed later in the ablation study, the full model better preserves prompt-specified semantics and avoids collapsing to dominant background textures, which is consistent with the FID and CLIP Score improvements reported in the quantitative results.

The generated samples also reflect prompt-level spatial cues, such as object placement and land-cover composition, despite the small and dense structures that are common in RS imagery. This indicates that UniEarth improves not only local appearance but also scene-level consistency with the input prompts, supporting the design intuition that a shared backbone with intermediate-level cross-modal anchoring enables reliable RS image synthesis under coarse text supervision.

D. Component Ablation

We first conduct an overall component ablation to assess the individual and combined contributions of CMA and DA-UCB within the unified framework. As shown in Table VI, vanilla unified training without either component provides the fully disabled baseline. Adding either CMA or DA-UCB improves this baseline, while combining both components achieves the best performance across all metrics. Removing the CMA module degrades generation quality. FID increases

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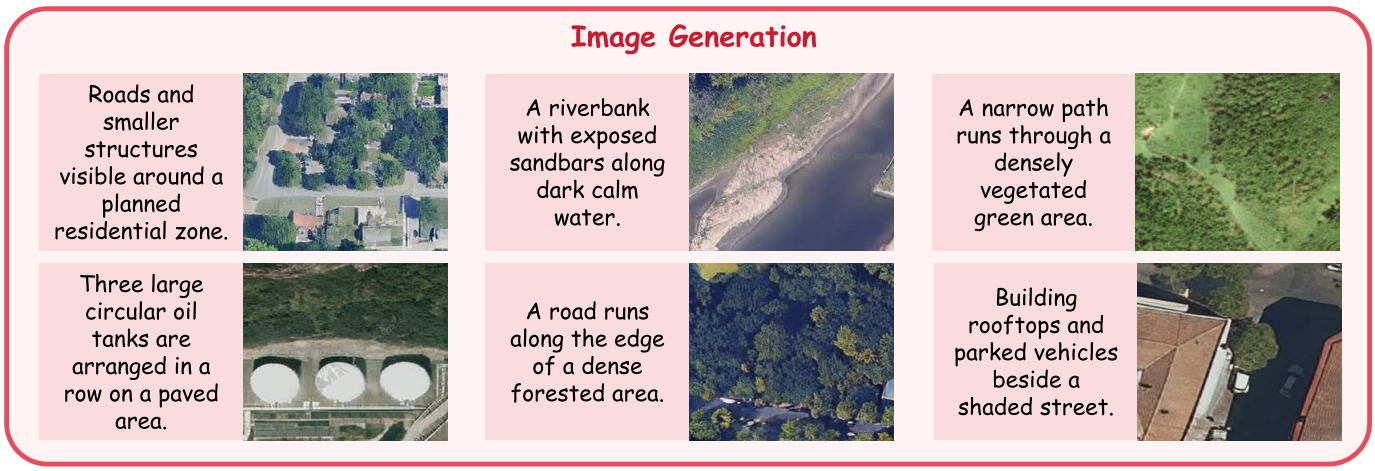


Fig. 5. Qualitative results for text-to-image generation. `UniEarth` generates photo-realistic RS images corresponding to diverse terrain and object descriptions.

TABLE VII
LAYER-WISE PAIRED–UNPAIRED COSINE-SIMILARITY GAPS OF RAW GLOBALLY MEAN-POOLED FEATURES WITHOUT CMA, EVALUATED ON 200 IMAGE–TEXT PAIRS

Layer m	1	3	12	24
Paired–unpaired gap	0.0027	0.0006	0.0006	0.0019

from 13.56 to 19.47, and the CLIP Score also drops, indicating that the T2I branch becomes more susceptible to texture-level fitting without intermediate-level semantic anchoring. The understanding metrics are likewise affected, suggesting that CMA benefits the shared representation used by both branches rather than improving generation alone. Disabling DA-UCB also consistently degrades performance: VQA Avg. Acc (All) decreases from 91.84% to 83.87%, FID deteriorates from 13.56 to 15.75, and the captioning metrics drop accordingly. This confirms that long-tail-aware difficulty allocation is essential for stable joint training. Together, the two components provide complementary improvements to both generation and understanding, and Sections VI-E and VI-F analyze each of them in more detail.

E. Analysis of CMA

1) Mechanism-Level Analysis: To provide mechanism-level evidence that CMA effectively establishes cross-modal semantic anchoring at the intermediate representation level, we conduct two complementary analyses on 200 image–text pairs sampled from the T2I corpus, namely cosine similarity distribution analysis and cross-modal retrieval evaluation.

Layer-Wise Raw-Feature Similarity Analysis: Using 200 image–text pairs, we examine the raw globally mean-pooled features at layers $m \in \{1, 3, 12, 24\}$. As shown in Table VII, without CMA, the mean paired–unpaired cosine-similarity gaps are only 0.0027, 0.0006, 0.0006, and 0.0019, respectively. Although the absolute similarity values vary with network depth, the paired and unpaired distributions remain highly overlapping across all examined layers.

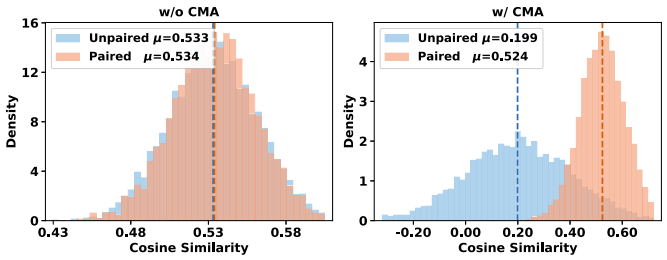


Fig. 6. Cosine similarity distributions between text and image intermediate features. (Left) Without CMA, paired and unpaired distributions largely overlap. (Right) With CMA, paired similarity is significantly higher than unpaired, demonstrating effective cross-modal alignment.

Cosine Similarity Distribution: We compare the representations used for evaluation in the two model variants. For the model without CMA, we use the raw globally mean-pooled text and image features extracted from the intermediate hidden states of the shared backbone. For the model with CMA, these pooled features are further mapped through the modality-specific text and image projectors, and the resulting projected representations are used. We then compare the cosine similarity distributions for matched and mismatched image–text pairs.

As shown in Fig. 6, for the model without CMA, the mean similarities of matched and mismatched pairs in the raw feature space are $\mu_{\text{paired}} = 0.534$ and $\mu_{\text{unpaired}} = 0.533$, respectively, yielding a gap of about 0.001. This near-zero gap indicates that the raw intermediate features do not form a discriminative cross-modal metric space. For the model with CMA, the corresponding mean similarities in the learned projection space are $\mu_{\text{paired}} = 0.524$ and $\mu_{\text{unpaired}} = 0.199$, yielding a substantially larger gap of 0.325. The clear separation between the two distributions shows that CMA learns a more discriminative representation space for cross-modal semantic matching.

Cross-Modal Retrieval: Using the same feature definitions as in the cosine similarity analysis, we construct the cross-modal similarity matrices from the raw globally mean-pooled intermediate features for the model without CMA and from

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TABLE VIII
CROSS-MODAL RETRIEVAL PERFORMANCE USING INTERMEDIATE-LAYER FEATURES. CMA IMPROVES RETRIEVAL RECALL COMPARED WITH THE MODEL WITHOUT CMA

Metric	w/o CMA (%)	w/ CMA (%)	Δ (%)
t2i R@1	0.5	**11.5**	+11.0
i2t R@1	0.5	**10.5**	+10.0
t2i R@5	2.0	**36.5**	+34.5
i2t R@5	2.5	**40.0**	+37.5
t2i R@10	6.0	**59.0**	+53.0
i2t R@10	4.5	**56.5**	+52.0

the projected representations for the model with CMA. For each text query, we rank all 200 images by cosine similarity and check whether the ground-truth image appears in the top-K results, denoted as text-to-image (T2I) retrieval. The reverse direction, image-to-text (i2t) retrieval, is evaluated analogously. Results are reported in Table VIII.

Without CMA, R@1 is 0.5% in both retrieval directions, matching the random baseline of $1/200 = 0.5\%$. This indicates that the raw, unprojected intermediate features of the model without CMA do not form a discriminative shared metric space. With CMA, retrieval in the learned projection space reaches R@1 values of 11.5% and 10.5% for T2I and i2t retrieval, respectively, while the corresponding R@10 values reach 59.0% and 56.5%. These results demonstrate that the learned CMA projection space provides substantially stronger cross-modal discrimination than the raw feature space available without CMA.

Together, these two analyses provide strong evidence that CMA functions as an effective semantic anchoring mechanism. It enforces meaningful cross-modal alignment at the intermediate representation level, which in turn stabilizes the generation branch under coarse RS annotations and benefits joint optimization across both tasks.

2) Comparison With Alignment Baselines: We further compare CMA with a simpler alignment baseline, namely mse alignment, to examine whether the observed gains mainly come from adding a generic cross-modal regularizer. For fair comparison, mse alignment is applied to the same intermediate text and image representations used by CMA, with the same projection layers, training schedule, and alignment weight. The only difference lies in the alignment objective: instead of bidirectional InfoNCE, mse alignment directly minimizes the L2 distance between paired text and image representations. Results are reported in Table IX.

As shown in Table IX, mse alignment improves over vanilla unified training, indicating that cross-modal regularization is beneficial. Nevertheless, CMA consistently achieves better generation and understanding performance than mse alignment. Unlike L2 alignment, the contrastive objective in CMA captures relative semantic structure and yields more discriminative cross-modal representations. This suggests that the gain is not merely due to adding a generic alignment loss, but depends on the intermediate-level contrastive semantic anchoring introduced by CMA.

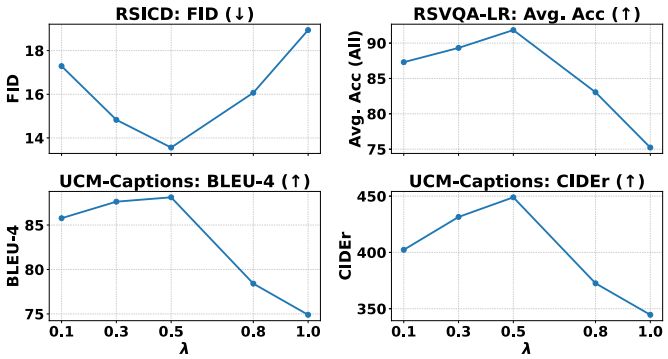


Fig. 7. Sensitivity analysis of the alignment weight λ. We report FID on RSICD, Avg. Acc (All) on RSVQA-LR, and BLEU-4/CIDEr on UCM-Captions.

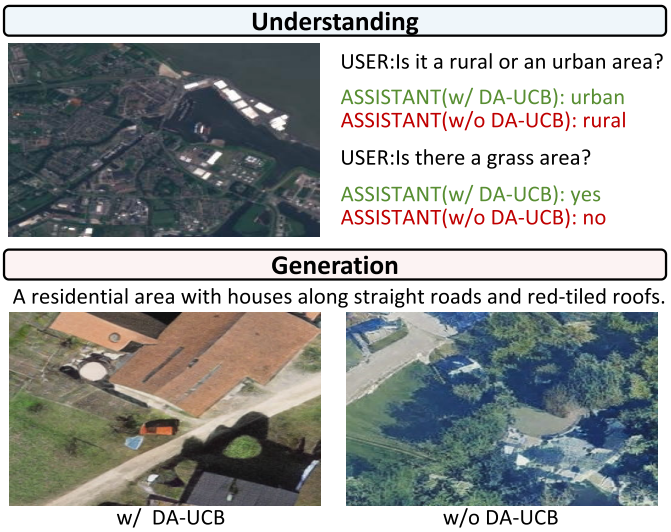


Fig. 8. Qualitative comparison of UniEarth with and without DA-UCB on representative hard samples. The examples include both understanding and generation cases.

3) Sensitivity to Alignment Weight λ: To validate our choice of the alignment weight λ, we conduct a sensitivity analysis by varying $\lambda \in \{0.1, 0.3, 0.5, 0.8, 1.0\}$ while keeping all other settings fixed. Results are reported in Fig. 7.

When $\lambda = 0.1$, the contrastive signal is weak, leading to insufficient cross-modal alignment and a higher FID of 17.29. Increasing λ to 0.5 improves both generation and understanding metrics, reaching the best FID of 13.56, RSVQA-LR Avg. Acc (All) of 91.84, BLEU-4 of 88.12, and CIDEr of 448.94. When λ is increased to 1.0, performance drops across all metrics, with FID deteriorating to 18.94 and RSVQA-LR Avg. Acc (All) decreasing to 75.23. These results indicate that a moderate alignment strength is important for stable joint optimization, allowing CMA to serve as a semantic regularizer without overwhelming task-specific learning objectives.

F. Analysis of DA-UCB

1) Qualitative Effect on Hard Samples: We further analyze the effect of DA-UCB on representative hard samples associated with low-frequency semantics. In our setting,

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TABLE IX
COMPARISON WITH ALIGNMENT AND IMBALANCE-AWARE BASELINES

Setting	RSICD		RSVQA-LR	UCM-Captions	
	FID↓	CLIP Score↑	Avg. Acc (All)↑	BLEU-4↑	CIDEr↑
Vanilla unified training	45.80	19.81	83.81	72.67	351.75
+ Focal Loss (replace DA-UCB)	17.63	22.84	69.18	60.13	294.71
+ MSE Alignment (replace CMA)	16.82	25.87	88.15	84.96	432.17
UniEarth (full)	**13.56**	**30.51**	**91.84**	**88.12**	**448.94**

underrepresented categories often manifest as persistently high-loss samples, since they are not sufficiently optimized under standard training. Fig. 8 presents qualitative comparisons on both generation and understanding tasks for such hard cases.

For visual question answering, we select questions associated with rare object categories or infrequent spatial configurations. Without DA-UCB, the model frequently produces answers biased toward dominant categories, consistent with a bias toward frequent categories under standard training. With DA-UCB, the model more accurately identifies the rare objects and answers accordingly, indicating improved recognition of underrepresented semantics. As shown in the understanding example, the model with DA-UCB answers both questions correctly, while the model without DA-UCB fails on both.

For image generation, we select prompts whose target semantics are easily overwhelmed by high-frequency concepts under standard training. Without DA-UCB, the model tends to collapse toward high-frequency background textures such as homogeneous vegetation or generic urban patterns, while failing to reflect the category-specific semantics specified by the prompt. With DA-UCB enabled, the generated images better match the target scene structure and foreground objects, suggesting improved modeling of underrepresented semantic patterns. In the example shown in Fig. 8, DA-UCB better preserves the prompt-specified residential layout and red-tiled roofs. Overall, by dynamically balancing exploration and exploitation, DA-UCB promotes the learning and generation of these long-tail semantic objects, improving both recognition and synthesis.

2) Comparison With Imbalance-Aware Baselines: We further compare DA-UCB with focal loss, a representative imbalance-aware baseline. Under the same backbone and training protocol, focal loss serves as a confidence-based reweighting strategy, whereas DA-UCB performs trajectory-based difficulty allocation during training. Results are reported in Table IX.

As shown in the table, focal loss improves FID from 45.80 to 17.63 compared with Vanilla unified training, but substantially degrades understanding performance, leading to an inferior overall tradeoff. One possible reason is that many RS understanding targets consist of short, low-entropy answer tokens, which can be predicted with high confidence early in training. Focal loss may therefore down-weight these tokens and reduce the effective supervision received by the understanding branch. In contrast, DA-UCB estimates sample difficulty from loss trajectories rather than from instantaneous token-level confidence, which helps emphasize under-explored semantic patterns during joint training while preserving stable supervision for low-entropy targets.

TABLE X
PARAMETER SENSITIVITY ANALYSIS OF THE HARD-SAMPLE WEIGHT ω_{HARD}

ω_{hard}	RSICD	RSVQA-LR	UCM-Captions	
	FID ↓	Avg. Acc (All) ↑	BLEU-4 ↑	CIDEr ↑
1.0	15.75	83.87	80.10	392.84
2.0	**13.56**	**91.84**	**88.12**	**448.94**
3.0	15.38	81.04	83.24	415.36
5.0	19.63	75.31	80.86	337.18

3) Sensitivity to Hard-Sample Weight ω_{Hard}: To validate the choice of ω_{hard} in (17), which controls the reweighting strength assigned to hard-arm clusters in DA-UCB, we conduct a sensitivity analysis by varying $\omega_{\text{hard}} \in \{1.0, 2.0, 3.0, 5.0\}$ while keeping all other hyperparameters fixed.

As shown in Table X, $\omega_{\text{hard}} = 1.0$ yields the weakest results among moderate settings, confirming that the dynamic reweighting mechanism is essential for handling long-tailed RS distributions. Increasing ω_{hard} to 2.0 achieves the best overall balance: FID reaches 13.56 on RSICD, VQA Avg. Acc (All) reaches 91.84% on RSVQA-LR, and CIDEr reaches 448.94 on UCM-Captions. When ω_{hard} is further raised to 3.0, both generation and understanding degrade, suggesting that over-emphasizing hard samples introduces noise from ambiguous or mislabeled instances. At the extreme value $\omega_{\text{hard}} = 5.0$, performance drops sharply across all metrics, even falling below the $\omega_{\text{hard}} = 1.0$ baseline, indicating that excessively large weights cause the model to overfit to noisy outliers. These results confirm that a moderate reweighting strength of $\omega_{\text{hard}} = 2.0$ best balances the exploration of rare categories with stable exploitation of well-learned patterns.

4) Class-Wise Analysis of DA-UCB: To examine the class-wise effect of DA-UCB, we conduct additional analyses on AID and NWPU-RESISC45 based on their class distributions. For AID, we group the classes according to the number of unique training images in the adopted 20% split. Classes with at least 74 training images are assigned to the head

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TABLE XI
CLASS-WISE ABLATION OF DA-UCB ON AID AND NWPU-RESISC45. RESULTS ARE REPORTED AS ACCURACIES (%), AND Δ DENOTES THE IMPROVEMENT IN PERCENTAGE POINTS

Dataset	Subset	w/o DA-UCB	w/ DA-UCB	Δ
AID	Head	96.90	**98.78**	+1.88
	Middle	90.80	**95.16**	+4.36
	Tail	87.90	**96.87**	+8.97
NWPU	Bottom-10	81.14	**90.57**	+9.43

Note: The AID group results and the NWPU Bottom-10 result are macro-averaged over classes. Bottom-10 refers to the ten lowest-performing classes under the setting without DA-UCB.

group and represent the many-shot classes. Classes with 60–72 training images are assigned to the middle group and represent the medium-shot classes. Classes with at most 58 training images are assigned to the tail group and represent the few-shot classes. The thresholds are determined by the natural frequency gaps in the training split. The resulting partition contains 10 head classes, 11 middle classes, and 9 tail classes.

NWPU-RESISC45 has a balanced training distribution, with the same number of training images in all 45 classes. A frequency-based division into head, middle, and tail groups is therefore not applicable. We report the macro-averaged accuracy of the ten lowest-performing classes identified under the setting without DA-UCB.

As shown in Table XI, DA-UCB improves all three AID frequency groups, with progressively larger gains as the training frequency decreases. The improvement increases from 1.88 percentage points on the head classes to 4.36 points on the middle classes and 8.97 points on the tail classes. The performance gap between the head and tail groups is consequently reduced from 9.00 percentage points without DA-UCB to 1.91 percentage points with DA-UCB. These results show that DA-UCB reduces the optimization bias toward frequent categories and improves the learning of underrepresented samples.

On NWPU-RESISC45, the macro-averaged accuracy of the ten lowest-performing classes identified under the setting without DA-UCB increases from 81.14% to 90.57%, corresponding to a gain of 9.43 percentage points. This result shows that DA-UCB also benefits visually ambiguous and persistently difficult categories when the class distribution is balanced.

G. Failure-Case Analysis

Fig. 9 shows representative limitations of UniEarth in both understanding and generation. In the image-captioning example, the model describes the residential neighborhood but overlooks the clearly visible ground track field, indicating incomplete semantic coverage in mixed scenes. In the first T2I example, the prompt requests a vertical road dividing the green fields, whereas the generated road is curved and does not satisfy the required spatial layout. In the second T2I example, the prompt refers to a concrete structure, but the model generates a small blue pool-like object in dense vegetation, showing incorrect grounding of object attributes and scene content. These cases suggest that, although UniEarth achieves strong overall performance, it can still struggle with localized semantic targets, precise spatial control, and fine-grained attribute binding.

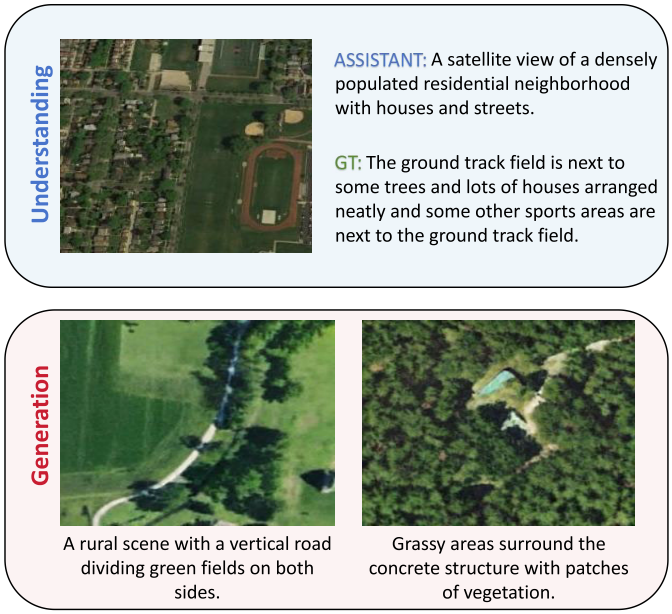


Fig. 9. Representative failure cases of UniEarth in image captioning and T2I generation, including semantic omission, inaccurate spatial layout, and incorrect object attributes.

VII. DISCUSSION AND LIMITATIONS

While UniEarth demonstrates strong performance across both understanding and generation benchmarks, several limitations remain and point to directions for future work.

Methodological Limitations: CMA uses globally mean-pooled representations at a fixed intermediate layer, which keeps the alignment lightweight but may not fully capture fine-grained object-level correspondences or adapt optimally across different tasks and annotation granularities. Its effectiveness also depends on the quality of paired image–text supervision. DA-UCB uses persistent training loss as a practical proxy for sample difficulty. Although effective for long-tailed data, noisy or ambiguous high-loss samples may receive excessive emphasis when the hard-sample weight is too large, as suggested by the results in Table X. DA-UCB also introduces several clustering and reweighting hyperparameters, and its benefit may be less pronounced on relatively balanced datasets. Future work will investigate region-aware alignment and noise-aware difficulty estimation.

Modality Coverage: The current framework is trained and evaluated exclusively on optical RS imagery. However, real-world EO relies on diverse sensor modalities, including SAR, hyperspectral, and infrared imagery, each of which exhibits distinct statistical properties and physical interpretations. Extending UniEarth to these modalities would require modality-aware tokenization for different signal characteristics and adapting CMA for cross-sensor alignment.

Scaling to Ultrahigh-Resolution Imagery: Extending UniEarth toward ultrahigh-resolution and gigapixel-scale overhead imagery represents a promising direction for

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future work, enabling applications such as wide-area small-object analysis and large-scene cartographic generation. The linear-complexity Mamba-2 backbone provides a favorable foundation for such extensions, and hierarchical or patch-wise scanning strategies tailored to overhead imagery can be naturally integrated to further enhance scalability.

Understanding Task Scope: The understanding capabilities of `UniEarth` are validated on closed-form tasks such as scene classification, short-answer VQA, and image captioning. More complex reasoning tasks, including open-ended visual dialog, spatial relationship reasoning, multitemporal change detection, and grounded localization, remain unexplored. Extending `UniEarth` to support these tasks would further test the generality of the unified next-token prediction framework in RS interpretation.

VIII. Conclusion

In this work, we propose `UniEarth`, a unified foundation model tailored for MMU and image generation in the RS domain. Extensive experiments on five benchmarks demonstrate that understanding and generation can be effectively integrated within a unified next-token prediction framework. The results further show that effective joint training requires both cross-modal semantic alignment and difficulty-aware optimization. CMA provides intermediate-level semantic anchoring between image and text representations, thereby improving generation under coarse RS annotations, while DA-UCB dynamically reallocates the learning focus toward persistently difficult samples under long-tailed data distributions. With a compact corpus of approximately one million RS samples, `UniEarth` achieves state-of-the-art results across all evaluated tasks, including a 44.6% relative reduction in FID, a 20.1% relative improvement in CIDEr, and leading accuracy on VQA and scene classification benchmarks. Overall, these findings demonstrate the effectiveness of jointly modeling RS understanding and generation and highlight the importance of addressing both cross-modal misalignment and data imbalance in unified training.

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